NIT ARUNACHAL PRADESH

ADDOVEDI

ENTER THE ARENA

OFFICIAL EVENT RULEBOOK

EDGE OF COLLAPSE

Design light. Build strong. Hold the load.

PART OF ADDOVEDI 2026

PRIZE POOL	TEAM SIZE	TYPE
₹8,000 cash	**3 to 4 members**	**Bridge model engineering**

Date, venue and schedule: to be announced

01 Event at a Glance

Item	Details
Event	Edge of Collapse
Type	Bridge model engineering
Team size	Minimum 3, maximum 4 members
Format	Design and Inspection, Knowledge of the Bridge, Load Test
Registration fee	₹50 per member (₹150 for 3 · ₹200 for 4)
Prize pool	₹8,000 cash (total worth ₹17,000 including goodies)
Date	To be announced
Venue	To be announced

Items marked To be announced are not yet confirmed. The organisers will announce them before the event.

02 Edge of Collapse

Design light. Build strong. Hold the load.

Introduction

Edge of Collapse is a structural design and fabrication competition that challenges participants to build a lightweight bridge model capable of carrying the maximum possible load. The event tests understanding of structural behaviour, load paths, truss geometry, joint quality and material efficiency.

Problem statement

Participants must design and build a three-hinged bridge model that spans a fixed gap and supports a progressively increasing load applied at mid-span. The winning bridge is the one that achieves the highest load-to-self-weight ratio while following all dimensional and material constraints.

Materials

A bridge may only be built from the following:

- Wooden ice-cream (popsicle) sticks of standard size, with no cutting of the stick cross-section beyond trimming to length.
- Water-based white adhesive (such as PVA / white glue). Quick-set, epoxy and hot-melt adhesives are not permitted.
- Cotton thread is permitted only for decorative binding and must not be a structural element.
- Metal, plastic, cardboard, bamboo, or any pre-made construction kit or pre-fabricated part is strictly prohibited.

Event structure

Round 1 – Design and Inspection

- Every bridge will be measured by the judges before testing.
- Judges assess the aesthetic looks of the bridge, including neatness of construction, joint quality and originality of the structural design.

- A bridge that does not meet the specifications will be disqualified directly.

Round 2 – Knowledge of the Bridge

- Round 2 is only for checking the team's practical knowledge.
- Teams will be questioned on the bridge they have built. Judges assess the team's knowledge of their own bridge.

Round 3 – Load Test

- Each team gets one official load test; the bridge is placed on two supports and loaded at mid-span. Load will be added gradually in equal increments through a loading block at the centre of the span.
- The test ends when the bridge fractures, a joint separates, or the mid-span deflection exceeds the limit set by the judges.
- The maximum load held before failure is recorded as the team's load capacity.

Judgement criteria

- Efficiency Ratio = Maximum load held / Self-weight of the bridge. The higher the ratio, the higher the rank.
- In case of a tie, the lighter bridge wins; if still tied, the bridge with lower deflection at the common load wins.
- A bridge that fails to span the gap, or collapses under the loading block weight itself, fails Round 3.

Bridge specification

Note: Any image shown in this rulebook is for illustration only. It does not depict the actual testing conditions.

Dimensions

- Overall length: 50 cm.
- Deck on which the car will run: 45 cm.
- Clear span between the supports during testing: 45 cm.
- The bridge must be a three-hinged bridge.
- Bridge must rest on the supports without any anchoring, nails or clamps.

Deck

- The bridge must have a continuous deck at least 5 cm wide so the loading block can be placed at the centre.
- Space must be available at mid-span for the load to be placed.
- Free passage of a 4 cm x 4 cm test block along the deck must be possible.

Event rules

Construction

- Bridges may be built in advance; teams must bring the finished model to the venue.
- No modification, repair or addition is allowed once the bridge has been weighed and submitted.
- Adhesive must be fully cured before submission. Wet or tacky bridges will not be accepted.

Safety

- Participants must stay behind the safety line during load testing.
- Any damage to testing equipment caused by misuse may lead to disqualification.

Consequences of violation

- Violation of any bridge rule, including use of prohibited materials or exceeding the size or weight limits, will result in immediate disqualification and ineligibility for associated prizes.

Contest rules

Submission

- Bridges must be submitted at the registration desk before the stated deadline on the day of the event.
- Each submitted bridge will be tagged with a team number and kept in the holding area until testing.

Time limits

- Each team gets 5 minutes at the testing station, including placement of the bridge and loading.
- The timer will run continuously once the team is called to the testing station.

Number of models

- Each team may submit only one bridge.

Team specifications and conduct

- Teams must maintain the spirit of healthy competition and must not damage other teams' models or equipment.
- Any participating candidate or team that does not meet the norms of the event will be disqualified.
- Judges reserve the right to disqualify any team involved in misconduct.
- In case of disputes or discrepancies, the decision of the organisers and judges will be final and binding.
- Organisers reserve the right to modify any or all rules. Any changes will be communicated to registered participants.

Prizes

Prize pool: ₹8,000 in cash, with total prize worth ₹17,000 including goodies.

03 Rules for Every Event

These rules apply to every ADDOVEDI 2026 event. The event rules in the previous section add to them.

- Follow all instructions from organisers, volunteers, coordinators and judges.
- Arrive on time. Late arrivals may lose build or round time.
- Plagiarism, copying or fraudulent methods are not allowed and lead to disqualification.
- Teams must keep to the spirit of healthy competition. They must not touch, damage or tamper with another team's work or equipment.
- Judges and organisers may disqualify any team involved in misconduct.
- In case of any dispute or discrepancy, the decision of the organisers and judges is final and binding.
- Organisers may modify any rule when needed. Changes will be communicated to registered participants.
- Eligibility, team size and registration details can differ by event. Check the event rules.

Registration fee

Team size	Fee per team (₹50 per member)
3 members	₹150
4 members	₹200

04 Contacts

Role	Details
Coordinator 1	Dikenno Pul: 6909125886
Coordinator 2	Mission Doley

WhatsApp group

Scan the QR code with the WhatsApp camera to join the group for all future updates.

Edge of Collapse

05 Details to Be Announced

The following details are not yet confirmed. The organisers will announce them before the event.

Detail	Status
Date, venue and clock time of the event	**To be announced**
Eligibility	**To be announced**
Mission Doley phone number	**To be announced**